



Drastic Changes Happening In T&M With Arrival Of 5G

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Deepshikha Shukla

When we travel by an aircraft or a ship, our safety relies on the operational accuracy and efficiency of these modes of transport. To ensure safety of travellers, aerospace and marine applications need high-end testing of every component in every possible condition. These are critical applications that need flexible and scalable test solutions to check their operational accuracy in harsh environmental conditions.

With the evolution of new technologies, changing standards and regulations, the global electronic test equipment market will undergo significant transformations. As per a Frost & Sullivan report, with the arrival of 5G, drastic shifts in signal generators, signal and spectrum analysers, network analysers and power meter segments of the electronic test and measurement (T&M) market are expected by 2020.

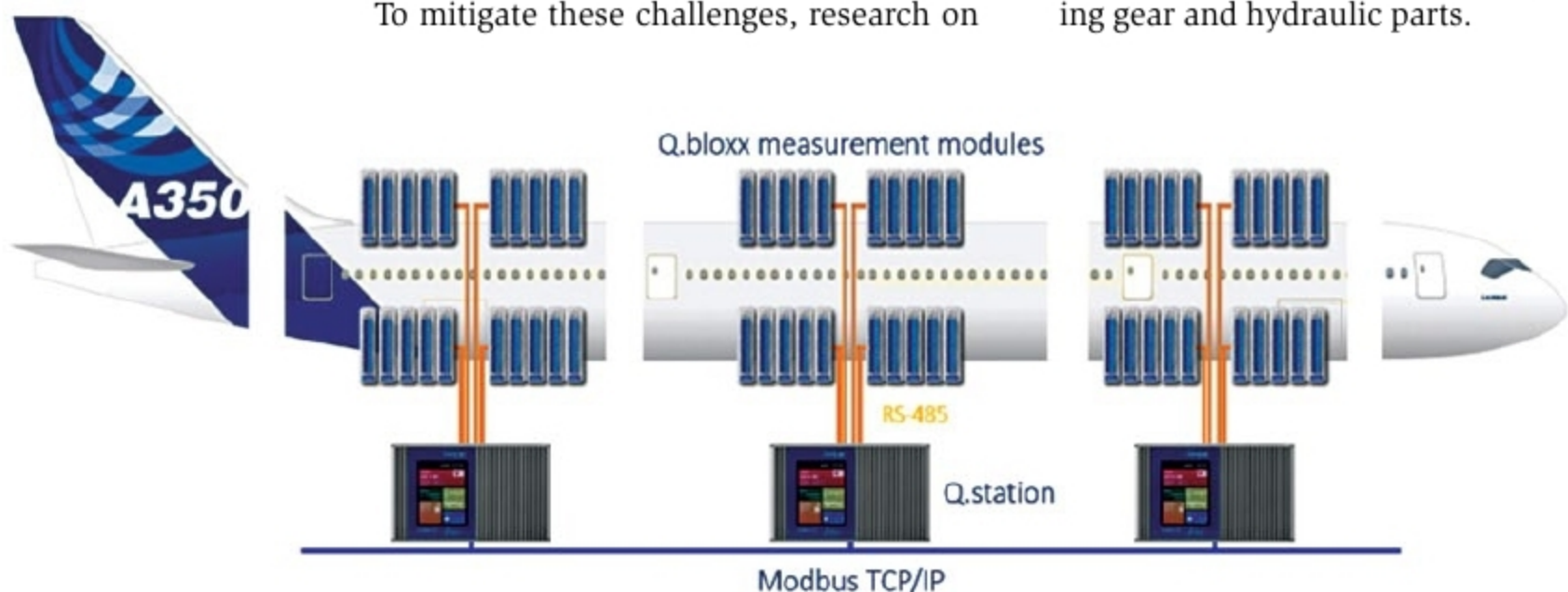
With increased capabilities and thus complexity of aircraft and ships, customers face challenges in testing, validation and verification of the modern equipment. To mitigate these challenges, research on

millimetre-wave frequencies (30GHz or above) for special-purpose radars and other equipment has been increased tremendously.

Aerospace

Different kinds of testing required for aerospace applications are flight, environmental, climatic, vibration, structural, instrumentation, fatigue, MIMO, wind tunnel, pressure measurement, lightning, gap measurement, air data, non-destructive, dynamics measurement, material, engine, corrosion, electrical systems, power supply, production and assembly. And yet the list may be incomplete! Following are a few examples of the latest in aerospace T&M equipment.

Gantner Instruments offers data acquisition systems for testing the strength of individual components and complete assemblies. This includes strain and force measurement, controlling and monitoring actuators, fatigue testing of turbine components, and testing of landing gear and hydraulic parts.



Q.station X (Credit: www.gantner-instruments.com)